# AI.311

## Robotics Planning

### COURSE OVERVIEW

Deliberative planning for robots: task, motion, and manipulation planning under constraints.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

# ORIENTATION: PLANNING TO ACT

**KEY TOPICS**

- Task vs. motion planning
- Why deliberate
- Planning in robotics
- Course roadmap

**TEACHING FOCUS**

Decide what to do before moving.

**KEY TAKEAWAY** Planning prevents costly mistakes.

ROBOTICS PLANNING

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SESSION

60 MINUTES

## TASK PLANNING

**KEY TOPICS**

- HTN and STRIPS planning
- Goal decomposition
- Plan execution and monitoring
- Replanning

**TEACHING FOCUS**

Structure complex missions.

**KEY TAKEAWAY** Task planning gives a robot purpose.

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SESSION

60 MINUTES

## MOTION PLANNING

**KEY TOPICS**

- Configuration space
- Roadmaps and PRM
- RRT and RRT*
- Optimal planning

**TEACHING FOCUS**

Find collision-free paths.

**KEY TAKEAWAY** Motion planning finds the way.

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SESSION

60 MINUTES

# MANIPULATION PLANNING

**KEY TOPICS**

- Grasp planning
- Pick-and-place
- Two-arm coordination
- Constrained manipulation

**TEACHING FOCUS**

Plan physical interaction with objects.

**KEY TAKEAWAY** Manipulation planning moves things intentionally.

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SESSION

60 MINUTES

# PLANNING UNDER UNCERTAINTY

**KEY TOPICS**

- Contingent planning
- Belief-space planning
- Online replanning
- Robust plans

**TEACHING FOCUS**

Plan when outcomes are uncertain.

**KEY TAKEAWAY** Good plans anticipate failure.

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SESSION

60 MINUTES

## ROBOTICS PLANNING & REVIEW

**KEY TOPICS**

- Planning demo on robot
- Course recap and assessment
- Toward social robotics (AI.312)
- Planning competition

**TEACHING FOCUS**

Execute a full task–motion plan.

**KEY TAKEAWAY** Plans turn goals into robot motion.